Chapter 1

Teaching subtitling in the times of generative AI

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This chapter explores the integration of Generative Artificial Intelligence (GenAI) into subtitler training and its impact on translator education. Drawing on teaching experiences and research insights, it presents strategies for designing and delivering training courses that prepare future subtitling professionals to work effectively in an AI-enhanced industry. Subtitling is a multifaceted translation practice that requires technical, linguistic and cultural skills. Recent developments in GenAI are reshaping how these skills are taught and applied. Based on realistic scenarios and exploratory teaching methods, the chapter examines how Large Language Models (LLMs) can support different stages of the subtitling process through inquisitive integration. The discussion encompasses practical approaches to integrating GenAI tools into the classroom, drawing on examples related to the translation of cultural references and template creation. The chapter adopts a hands-on problem-solving approach to training that encourages students to evaluate technological possibilities whilst developing foundational knowledge. Through examples from teaching practice, it shows how comparing different AI solutions and assessing their suitability for specific tasks helps students make informed decisions about implementing automated solutions. This approach positions students as active agents in their learning process while helping them understand the potential and limitations of automation. Critically examining the role of educators in this changing landscape, the chapter advocates for training that prepares adaptable professionals who can navigate technological developments whilst maintaining high standards and advocating for sustainable working conditions. More broadly, it contributes to discussions about providing students with the necessary tools and knowledge to shape sustainable careers in an increasingly automated media localisation industry.

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1 Introduction

This chapter addresses the question of integrating Generative Artificial Intelligence (GenAI) into subtitler training. Audiovisual translation (AVT) is a technology-driven field, and the reconfiguration of subtitling processes aptly encapsulates how accelerated GenAI-influenced changes affect education.

Subtitlers require technical, linguistic and transfer skills, as well as market awareness, to operate efficiently in the industry. Training professional subtitlers who can engage effectively with the industry involves "taking into account the linguacultural dimension as well as the technological possibilities and the market reality" (**Díaz-CintasRemael2021**: 62). Until relatively recently, technological tools in subtitling workflows focused on technical aspects such as synchronisation and segmentation, and sometimes automatic speech recognition (ASR) (**Georgakopoulou2020**). The field is experiencing exponential growth in the implementation of machine translation (MT) and GenAI solutions (**Slator2024**), pushed forward by Language Service Companies (LSCs) to speed up processes and address what they have called a *talent crunch* (**Estopace2017**; **Papercup2022**). The expansion of the media localisation industry has placed subtitling as one of the most active sectors of the market, resulting in growing interest from LSCs, increasing demands on production networks and pressure on the workforce.

This rapid transformation presents unique challenges for subtitling education. Both trainers and trainees must continuously adapt their knowledge and skills as technologies evolve. The traditional approach of mastering established tools and workflows needs to give way to developing adaptability and critical evaluation skills for emerging technologies. A survey conducted by **Slator2024** showed that different types of captioning and subtitling services involving AI are commonly offered by Language Service Providers.

Figure 1: . Results of a survey by Slator (**May2024**) that asked 129 Language Service Providers which language AI services their company offers

This chapter explores how we can structure subtitling training to address these challenges while maintaining high professional standards. Drawing on teaching experiences and research insights, I present strategies for designing and delivering training courses that empower future subtitling professionals to work effectively in an AI-shaped industry. Understanding the conditions, constraints and opportunities at every stage allows future professionals to make informed decisions and supports their communication and negotiation skills. The discussion encompasses practical approaches to integrating GenAI tools into the classroom within a project-based approach (**Mitchell-Schuitevoerder2020**), exploring conversational Large Language Models (LLMs), and implementing problem-solving activities to develop critical evaluation skills.

The chapter adopts a hands-on problem-solving approach developed through teaching practice, equipping future graduates to respond to the rapidly changing demands of the media localisation industry. The activities used as examples intentionally focus on non-conventional subtitler tasks to demonstrate the wide-ranging application of AI technologies and encourage a broader reflection on AI integration. The chapter examines how educators can keep up with technological developments and guide students to harness AI's potential while fostering a deep understanding of linguistic features, cultural contexts, and ethical considerations intrinsic to subtitling, automation, and the media industry. More broadly, this chapter contributes to discussions about providing students with the necessary tools and knowledge to shape a sustainable career.

2 Evolving landscape of subtitling training

Training subtitlers requires addressing the technical requirements of the practice related to synchronisation and spotting while balancing the linguistic and cultural requirements. Traditional subtitling workflows for single language pairs followed a relatively straightforward process: transcription of the source content, translation, synchronisation (or spotting) of subtitles with the audiovisual content, and quality control (**Díaz-CintasRemael2021**). Each stage required specific skills and was typically carried out sequentially, with subtitlers working on standalone workstations using subtitling software.

The globalisation of the media industry has led to the implementation of template-based workflows (**Georgakopoulou2019**). In this model, LSCs create a master file, usually in English, which serves as the basis for pivot translations into multiple target languages (Figure **??**) and streamlines the multilingual subtitling process (**ValdezEtAl2023**). Templates are essential for the multilingual operations of transnational platforms. Netflix defines subtitle templates as "an edited, positioned, researched, annotated and checked subtitle file, timed to shot and audio, matching the source language of the associated content" (n.d.). These templates include not only the text but also technical parameters and notes that support translations into multiple languages. For example, in producing subtitles for Netflix's successful Korean show "Squid Game", an English template serves as the pivot for translations into multiple languages. A template for this show would include not only the script and timing but also essential cultural annotations. Critical elements that would require annotation include Korean honorific forms (indicating relationships between characters), cultural references (such as the children's games that structure the plot), and social hierarchies expressed through language choice. For instance, the relationship between the two characters who share a childhood friendship requires careful annotation as their use of informal language reflects their familiarity, while their interactions with other characters follow strict social hierarchies reflected in formal language patterns. These annotations in the English template help translators working into other languages understand and preserve these nuanced relationships and cultural elements in their target versions.

In some cases, these templates are *locked* and do not allow translators to alter the timings of the subtitles. Thus, subsequent translations need to adapt to the synchronisation and reading speed of the pivot language. This approach streamlines the production process but also transforms the subtitler's role from having control over technical and linguistic aspects to focusing primarily on translation within pre-established constraints.

Figure **??**. Template-based translation process

The subtitling landscape is experiencing an unprecedented transformation. As **SzarkowskaJankowska2024** observe, workflows are becoming increasingly automated. They suggest that manual spotting and transcription may soon become obsolete, replaced by automatic speech-recognition-based spotting. This coincides with Bolaños García-Escribano's views that these changes are "perhaps leading to scenarios where spotting will be increasingly automatised and linguists will serve as language engineers" (2025: 19). In current post-editing processes, AI tools in the form of ASR and MT reshape traditional practices and embed revision stages to accommodate the human verification of the automatic output (Figure **??**). This transformation is further accelerated by the shift from standalone workstations to cloud-based environments that integrate translation memories, quality control automation and MT and transfer the control of these resources to the LSCs.

The implementation of AI tools for technical aspects has a long tradition in

Figure **??**. Post-editing production processing using ASR and MT

subtitling software and subtitlers tend to be more acquainted with them. Features such as importing and segmenting scripts, automatic timing and synchronisation, shot-change identification and automated QC are standard in subtitling workflows. Well-established tools, such as Ooona, EZTitles and ZooSubs integrate these by default, with some even allowing customisation.

Core competencies in subtitling foster the mastering of technical aspects such as synchronisation and segmentation, while developing the linguistic and cultural awareness needed to produce high-quality translations. However, the current industry landscape demands additional skills. Subtitlers now need to understand and critically assess the output of multiple automated systems, from ASR to MT, when engaging with post-editing workflows. Post-editing AVT requires professionals to coordinate an increasing number of sources of information and evaluate them before deciding to keep, edit or fully replace automatically generated content.

The industry's push towards automation responds to multiple factors (Orrego-Carmona Forthcoming). The expansion of streaming platforms has increased the demand for subtitled content, with companies requiring fast turnaround times for multiple language combinations. LSCs argue that a talent shortage constrains their ability to meet these growing demands, leading them to implement hybrid and automated solutions (Iyuno SDI **Group2022**; **Marking2022**). However, professional associations disagree with this assessment, denouncing that low payment and poor working conditions, rather than a talent shortage, are the primary concerns (**AVTE2023**).

The changes in the production processes require a substantial revision of how we approach subtitler training. The focus must shift from teaching specific technical and translation skills to developing adaptable professionals who can critically engage with evolving technologies while maintaining high-quality standards. Like in other areas of translator training, programmes need to address new technological requirements (ASR, MT, automated spotting, LLMs), industry demands, critical assessment skills for AI-generated outputs, and professional development needs in a rapidly changing environment.

The challenge for training programmes is to balance translation skills and technical proficiency with critical awareness. Students need to understand not only how to use new tools but also how to evaluate their suitability for different projects, markets and contexts. This involves developing what **Tipton2024** calls digital reflexivity: the ability to identify challenges in the external environment, evaluate available resources and implement appropriate strategies. Such an approach helps students navigate the changing landscape while maintaining professional standards and developing sustainable careers.

3 Designing AI-enhanced training and digital reflexivity

Digital reflexivity refers to students' ability to evaluate challenges and resources, and implement solutions. This can be illustrated through subtitling project work where students must make informed decisions at every stage. For instance, when creating subtitles, students first analyse their source product's translation needs, including the technical, cinematographic, linguistic and cultural aspects. They then evaluate available tools and resources, from subtitling software to AI solutions, justifying how these align with project requirements. Through this process, students learn to identify potential challenges in the external environment (such as technical constraints or cultural references), assess their resources (both technological and knowledge-based), and implement appropriate strategies to bridge any gaps.

This reflective practice manifests in practical decisions such as:

- Selecting appropriate software based on project needs
- Deciding which stages of the workflow can benefit from automation
- Evaluating when to use AI tools and when human expertise is crucial
- Documenting and justifying translation decisions

The resulting commentary demonstrates how students develop digital reflexivity by connecting theoretical knowledge with practical implementation whilst maintaining professional standards.

Training future subtitling professionals requires a comprehensive approach that recognises this multifaceted nature of the profession and the impact of technological developments on established practices. Encouraging students to explore technical, translation and technological aspects while examining the potential integration of automation into workflows should provide them with opportunities to develop their skills and systematically evaluate the performance of LLMs, ASR, MT and other AI systems. As an example of this type of reflective practice, this section focuses on the use of LLMs for the translation of cultural references to show how subtitling processes can be augmented through the critical use of automation tools (See O'**Brien2024** on augmentation in translation).

3.1 Contrasting and assessing translations for subtitles

Conversational LLMs and other AI tools can be integrated early on in the training process. A common exercise used in introductory subtitling classes is to share examples of cultural adaptations trainers and students find in the subtitles they encounter. After selecting some examples, students reflect on the aspects relevant to the translation of cultural references in subtitling (See **Díaz-CintasRemael2021**). These examples can be analysed using LLMs to allow students to assess subtitling decisions and understand the adaptation process, contrasting their developing knowledge of subtitling against the decisions of other subtitlers and the explanations provided by LLMs.

The example (**??**) below shows the transcript of the subtitles from *Anatomy of a Fall* (dir. Triet, 2023) shown on Amazon Prime in the UK with the original film, which includes dialogues in English, French and German.

- *Anatomy of a Fall*, 00:39:08:00-00:39:25:00
  - French:

Trois éléments en particulier : un rapport d'expertise sur des projections de sang de M. Maleski retrouvées au domicile du couple, et dont la direction semble indiquer qu'il a reçu un coup violent à la tête alors qu'il se trouvait sur le balcon du 2ème étage.

- – English:

Three elements in particular. Analyses of Mr Maleski's blood spatter found at the couple's domicile indicate he may have received a blunt force blow to the head when he was on the third-floor balcony.

Having access to the source and target versions of the text allows students to assess how LLMs could respond to the task. Students can use LLMs as agents to assess how the systems would respond to the tasks and how subtitling, as a practice, is depicted in the output. Example (**??**) shows the prompt used and the responses provided by three LLMs.

- *Prompt:* You are acting as an agent supporting the task of translating subtitles. This text is part of the script of the film Anatomy of a Fall. Translate it and adapt it to subtitles for the original film in English: Trois éléments en particulier : un rapport d'expertise sur des projections de sang de M. Maleski retrouvées au domicile du couple, et dont la direction semble indiquer qu'il a reçu un coup violent à la tête alors qu'il se trouvait sur le balcon du 2ème étage.

– ChatGPT (4o mini):

Subtitle Adaptation:

"Three key elements: an expert report on blood spatter belonging to Mr. Maleski found in the couple's home."

"The blood pattern suggests he was struck violently on the head while on the second-floor balcony."

Let me know if this fits well or if any adjustments are needed!

- – Copilot, GPT4:

Sure, here's the translation and adaptation for subtitles:

"Three key elements: an expert report on Mr. Maleski's blood splatter found at the couple's home, indicating he was struck violently on the head while on the second-floor balcony."

Does this work for your needs?

- – Claude 3.5 Sonnet:

Let *me help you with the subtitle translation and adaptation, focusing on clarity and proper subtitle format. Since this appears to be part of evidence being discussed in the film, I'll format it appropriately for subtitling:*

(1) Three elements in particular:
a forensic report on Mr. Maleski's blood spatter
found in the couple's home,
where the direction appears to indicate
he received a violent blow to the head
while on the second-floor balcony.

The text is broken down into readable chunks, each line keeping under the standard 42-character limit for subtitles. This maintains readability and follows standard subtitle timing practices. Would you like me to adjust the line breaks or pacing differently?

The results of this initial test indicate disparities across the systems. Apart from providing a translation, Claude (2,c.) provides context both about the scene and the nature of subtitling. The resulting translation also displays the limitation of subtitles (segmentation, character limitation, timing and readability), providing students with an opportunity to reflect on subtitling guidelines and instructions.

Having access to the official translation, students can ask the systems to compare the two versions. When asked to contrast the original translation against the new proposed translation, Claude again stands out by displaying contextual awareness (**??**). For instance, it states that the original version's decision to translate *domicile* as 'domicile' instead of 'home' reflects language that is 'more formal/legal, fitting for a courtroom scene'.

- *Prompt:* The translation of the subtitles in English reads as below. Can you explain the differences? from your translation? "Three elements in particular. Analyses of Mr Maleski's blood spatter found at the couple's domicile indicate he may have received a blunt force blow to the head when he was on the third-floor balcony."
  - Claude 3.5 Sonnet (excerpt):

5. Description of Injury:

- *My version: "violent blow"*
- *Official: "blunt force blow" (more precise medical/forensic terminology)*

6. Three key points: a forensic report on blood spatter from Mr. Maleski found at the couple's home, indicating he received a violent blow to the head while on the second-floor balcony. Floor Reference:

- *My version: "second-floor" (direct translation from "2ème étage")*
- *Official: "third-floor" (This reflects the difference in floor numbering between French and American systems - in France, the ground floor is "rez-de-chaussée," making "2ème étage" equivalent to the third floor in American English)*

The official translation is more appropriate for the legal/courtroom context of "Anatomy of a Fall," using more precise legal and forensic terminology while maintaining appropriate ambiguity for evidence being presented.

The integration of LLMs into subtitling training offers unique opportunities for students to actively engage with and validate the knowledge they acquire from traditional textbooks and guidelines, such as Netflix's. By independently comparing LLM outputs with established subtitling principles, students can develop their critical thinking skills and deepen their understanding of subtitling conventions. A clear example of this emerged when testing different LLMs with the translation of *deuxième étage*. All LLMs failed to correctly render the French floor numbering system into English, producing a literal 'second floor' instead of the culturally appropriate 'third floor'. This apparent limitation became a valuable teaching moment directly showing students why they need to be critical of the system, but also allowing them to learn about the subtitling of cultural differences. When presented with the official subtitled version, Claude provided the cultural and linguistic reasons behind this difference, highlighting the French convention where *rez-de-chaussée* is the ground floor, making *deuxième étage* equivalent to the third floor in the US English-speaking context. The output by ChatGPT, instead, stated that 'This change in floor number is critical. It could be a factual error or a discrepancy based on interpretation.' This interaction demonstrates how LLMs can support learning not just through their successes, but also through their limitations, helping students understand the complexity of cultural adaptation in subtitling.

3.2 Creating templates with cultural annotations

Template-based workflows are enhanced by including cultural annotations that support translators working in multiple target languages. As these workflows become more common in the industry, the creation and QC of templates also become more needed. For instance, recent years have seen the emergence of *template creators* as essential in subtitling industry workflows (**OziemblewskaSzarkowska2022**).

Comprehensive cultural annotations in pivot templates can reduce the research burden on translators (**ValdezEtAl2023**) and support the development of the industry. LLMs can be leveraged to help students understand the process of creating and working with annotated templates.[1] By prompting LLMs to identify and explain cultural references, students can develop skills in recognising cultural elements that require annotation and understanding the importance of clear, useful notes for other translators. This exercise helps them appreciate the dual role many subtitlers now play, translating content and facilitating translations into other languages through thoughtful documentation.

The example below includes an excerpt from the show *Derry Girls* (McGee, 2018-2022). The prompt mentioned the show and the task, as a way to narrow down the options and emphasise to students the relevance of a translation brief.

- *Prompt*: You are acting as an agent supporting the task of creating a subtitle template for the show Derry Girls. Using the transcript and dialogue list

[1] With thanks to Prof Agnieszka Szarkowska, who shared this idea and the materials she used to test it.

below, create annotations for a template explaining all cultural references that might be problematic for non-English native speakers and translators. List all references together with a brief one-sentence annotation.

Motherfuckers!
”Motherfucker” is my new thing.
Watched this film last night.
My dad got it off Pyro Pauline,
and it’s about these two lads,
and they wear these cracking suits
and they rock about,
just shooting people and eating cheeseburgers,
and they’re all,
”Motherfucker this, motherfucker that.”
It’s got your man in it.
What do you call him?
The disco dancer from Look Who’s Talking.

- – Claude 3.5 Sonnet

DERRY GIRLS SUBTITLE TEMPLATE - CULTURAL REFERENCES ANNOTATION

CONTEXT: Scene takes place in Northern Ireland (Derry/Londonderry) in the 1990s. CULTURAL/LINGUISTIC REFERENCES:

1. ”Motherfucker” - Reference to Pulp Fiction’s signature dialogue, showing 1990s American film influence on Irish youth culture.

2. ”Pyro Pauline” - Refers to a person dealing in pirated films, common in 1990s Northern Ireland.

3. ”lads” - Irish/British informal term for young men or boys.

4. ”cracking suits” - Northern Irish slang meaning excellent or impressive suits.

5. ”rock about” - Northern Irish vernacular meaning to walk around or hang out.

6. ”your man” - Irish/Northern Irish way of saying ”that guy” (not possessive).

7. ”disco dancer” - Reference to John Travolta, connecting his roles in Saturday Night Fever and Look Who’s Talking.

TRANSLATION NOTES:

- *Maintain informal teen speech register*
- *”Motherfucker” requires careful cultural adaptation*

- *Consider equivalent regional dialect markers in target language*
- *Film references should remain unchanged as proper nouns*

Would you like me to expand on any of these annotations?

- – ChatGPT 4o mini

Here's a list of cultural references from the provided dialogue in Derry Girls, along with annotations to assist non-native English speakers and translators:

*1. **"Motherfuckers!":**A strong expletive often used for emphasis; common in informal American English but can be considered vulgar, so care is needed in translation.*

*2. **"Motherfucker" is my new thing:**The phrase indicates adopting a bold or rebellious attitude, highlighting the character's enthusiasm for using strong language; understanding its connotation is important for context.*

*3. **"Watched this film last night. My dad got it off Pyro Pauline":** Refers to a film obtained informally, possibly through a local connection; "Pyro Pauline" may suggest a character known for being fiery or reckless, and understanding this local slang can enhance clarity.*

*4. **"Two lads":**"Lads" is a colloquial term in Irish and British English for boys or young men; it conveys camaraderie and familiarity, which may be less common in other cultures.*

*5. **"Cracking suits":**"Cracking" is British slang meaning excellent or impressive; this expression highlights the characters' stylishness and may not have a direct equivalent in other languages.*

*6. **"Shooting people and eating cheeseburgers":**A typical depiction of violence in action films juxtaposed with American fast food culture; may require additional context for audiences unfamiliar with these themes.*

*7. **"Your man":**An informal Irish phrase referring to a man whose name may not be known; this colloquialism can be confusing for those not familiar with regional dialects.*

*8. **"Look Who's Talking":**A popular 1989 romantic comedy featuring a talking baby; knowledge of this film is necessary to understand the reference and its cultural impact in the U.S.*

*9. **"The disco dancer":**This likely refers to a character known for dancing, reflecting a fun or upbeat personality; it might need context for audiences unfamiliar with the associated cultural style.*

These annotations can help non-native speakers and translators navigate the cultural nuances and humor present in the dialogue of Derry Girls.

Claude structures its response as a list of references and translation notes, while ChatGPT provides a narrative explanation of each reference. Claude's systematic approach, including the contextualisation of the scene in 1990s Northern Ireland, demonstrates a consideration of the broader cultural and historical setting essential for template creation. However, ChatGPT offers more detailed explanations of individual references, which could be particularly useful for students and translators unfamiliar with British and Irish culture.

Contrasting outputs from different LLMs is essential for developing students' critical skills. For instance, both systems correctly identify "your man" as an Irish expression, but their explanations differ in depth and utility for translators. Claude succinctly notes it an "Irish/Northern Irish way of saying 'that guy'", while ChatGPT elaborates on potential confusion for those unfamiliar with regional dialects. Such comparisons help students understand the importance of comprehensive yet clear cultural annotations.

The exercise also reveals potential limitations of LLMs. In earlier iterations of these prompts, some systems referenced DVDs, an anachronistic reference for the show set in the 1990s. This highlights the importance of holistic awareness in cultural annotation and the need for human verification. Students need to develop the skills and knowledge to critically evaluate LLM outputs, understanding that while these tools can support template creation, they require close monitoring and contextual understanding. Allowing them to compare these disparities emphasises their role and the need for them to sharpen their skills as they are directly confronted with the need to access their own knowledge and repertoire to evaluate the LLM output.

Moreover, the differences across outputs and details offered demonstrate how various LLM outputs can complement each other in the learning process. Students can appreciate how different annotation styles might serve different purposes in template creation and translation, by extension. Activities of this exploratory nature help develop their own strategies for creating clear, useful annotations that support multilingual subtitle production, and to evaluate those annotations produced by others.

3.3 Further integration into subtitling workflows

The examples above demonstrate how subtitling training can be enhanced by integrating AI tools to support different stages of the translation workflow while making students aware of the potential and impact of LLMs. Every step in the subtitling process covered in the training process could involve some degree of automation, allowing students to use their own AI agent to support skills they might be lacking. Placing the students in the driving seat and providing them with opportunities and abilities to make informed decisions will prepare them to face similar decisions in their future careers. This integration follows the approach of augmented translation; defined by CSA Research as a type of translation that follows "a technology-centric approach to amplifying the capabilities of human translators" (**Lommel2020**).

AI systems can support students throughout the subtitling process. This is in line with O'Brien's proposal for human-centred augmented translation that benefits translators at an individual level by enhancing their cognitive capabilities, "making a person more efficient, effective, creative, more competitive, more fulfilled" (2024: 402). In the pre-translation analysis, LLMs can help identify challenging cultural references and understand register variations. This initial exploration allows students to approach cultural transfer systematically, considering multiple options before making informed decisions.

When it comes to technical aspects, LLMs can suggest subtitle segmentation alternatives while maintaining students' agency in defining reading speeds, higher-level guidelines and character limitations (**SzarkowskaJankowska2024**). This helps them understand the principles behind technical parameters and encourages them to experiment with different solutions within the established constraints. Technical aspects could be extracted from guidelines and evaluated automatically using LLMs.

Quality control and documentation offer additional opportunities for the integration of AI tools in self-assessment. Students can use LLMs to identify inconsistencies and evaluate the potential reception of their decisions. They can use the system to extract records of their choices, evaluate consistency, and gauge the rationale behind their translations as decoded by the system. This can prove particularly useful for team projects, where groups need to coordinate the translation of multiple episodes. Working together, students can learn to develop strategies that ensure consistency across episodes, preparing them for real-world scenarios where large teams collaborate in localisation projects.

Recent developments in media localisation and AI point to further changes in how subtitles are created and integrated into audiovisual content (**Slator2024**). The emergence of LLMs capable of processing multimodal input will likely continue transforming subtitling workflows. In the near future, it is possible these systems will efficiently identify speakers, analyse context based on visual elements and suggest timing based on scene composition. Some of these tools are in development but not yet deployed. Even if these technologies are not currently integrated into media localisation processes at scale, preparing students to understand upcoming multimodal developments is essential. The expansion of accessibility services, such as audio description, and the development of automatic dubbing suggest professionals will still need to analyse interactions between different meaning-making modes and make decisions considering their target audiences. Activities and courses that support multimodal analysis and integrate AI tools while maintaining human agency in decision-making encourage future professionals to think strategically about subtitling workflows and their role in an evolving industry.

4 Tailoring workflows in subtitling training

The proposed integration of AI tools into subtitling training moves beyond teaching students how to use specific systems to promote an understanding of the considerations relevant to the products, processes and producers. It fosters a problem-solving mindset that enables students to develop flexible, context-appropriate and industry-informed workflows. By encouraging students to critically engage with different tools and approaches, they can develop the foundational knowledge and adaptability needed in the current industry landscape. Further, the continuous reference to academic knowledge and industry expertise empowers students to move swiftly between sectors.

A key component of this approach is allowing students to design their own subtitling projects and take ownership of all project-relevant decisions: from material selection to the tailoring of the translation situation. Adopting a project-based approach to translator training (**Kiraly2006**; **Mitchell-Schuitevoerder2020**) and rather than working with pre-defined parameters, students select the audiovisual products to be translated, define the production conditions, select the most suitable guidelines and reflect on the reception requirements that frame their translations. This autonomous decision-making process helps students understand how different factors—audience needs, technical constraints, cultural considerations, and technological tools—influence workflow design. For instance, a student might choose to subtitle a documentary aimed at a specialist audience, requiring careful consideration of terminology management and potentially benefiting from AI tools for quality assurance. Another might work on a comedy clip where cultural references and wordplay demand more human intervention and support for creative solutions. Classes can also integrate spaces to share and discuss the students' decisions along the way: short progress presentations or online discussion forums to provide progress updates can help students learn from each other.

Through these self-directed projects, students learn to:

- Evaluate which stages of the workflow can benefit from automation
- Assess the suitability of different AI tools for specific tasks
- Understand when human expertise is most critical and when automation provides the most advantages
- Translate their products, reflecting on specific translation needs
- Design quality control processes that combine automated and manual checks
- Document their decision-making process for future reference

The flexibility to explore different workflows bridges academic expertise with industry developments in meaningful ways. While academic approaches emphasise systematic analysis and an in-depth understanding of subtitling principles, industry practices focus on efficiency, market relevance and technological innovation. By allowing students to design their workflows, this independent project-based approach creates opportunities for them to understand how theoretical principles support informed decision-making in practical scenarios. Students learn to apply academic research on reading speeds (Kruger, **WisniewskaLiao2022**; **SzarkowskaBogucka2019**; **SzarkowskaEtAl2024**) and segmentation (Gerber-Morón, **SzarkowskaWoll2018**) to evaluate the output of automated systems, or draw on translation theory to assess MT suggestions. This integration of perspectives helps students develop digital reflexivity (**Tipton2024**) to support industry practices. As LSCs continue to deploy new automated solutions that challenge traditional workflows and companies develop more tools, students need to engage critically with the process, resort to solid foundational knowledge and be able to swiftly integrate into new technological production environments. Having a sound theoretical basis allows students to critically assess these developments and adapt their practice accordingly rather than following prescribed procedures.

This proposal promotes collaborative learning environments that benefit both students and trainers. When students are empowered to test different tools and assess their performance, they develop the confidence to access new knowledge autonomously. Rather than positioning tutors as the sole source of information, the responsibility of engaging with emerging technologies and industry practices is shared between tutors and students (**Kiraly2003**; **Kiraly2006**). This approach recognises that trainers themselves are navigating a rapidly evolving landscape and need to continuously update their skills. The resulting dynamic supports trainers' continuing professional development as they learn alongside students, fostering an environment of mutual growth that better reflects the constant technological changes in the industry.

5 Ethical considerations

The implementation of automated solutions and, in particular, the popularisation of LLMs has made discussions about the ethical dimensions of these practices essential (**Moorkens2020**; **Moorkens2024**). According to the Statement on Generative Artificial Intelligence published by AVTE, AudioVisual Translators Europe, the European Federation of National Associations of Audiovisual Translators, the use of AI in creative translation does not respond to creative needs but an intention to increase profit "while compromising the livelihood of professional translators and offering audiences subpar products" (2024: 5).

Training activities must be designed and implemented with careful consideration of ethical implications. Acknowledging that engaging with AI tools is now essential for employability in the industry, but it is also important to recognise and address the ethical and social issues of LLMs. Firstly, the use of copyright-protected materials in LLM training raises concerns about intellectual property rights. Creative industries, such as audiovisual translation, have been particularly active in denouncing the deployment of LLMs as an exploitative practice whose impact is incremented by the deployment of AI-powered production models (**AVTE2024**; **ATRAE2021**; En chair et en **Os2024**). Secondly, and more importantly, training should acknowledge how the implementation of AI tools might affect working conditions in the industry (**AVTE2023**; **KarakantaEtAl2022**; **KoponenEtAl2020**). Professional associations have pointed out that the push for post-editing workflows involving AST, MT and LLMs comes with a demand for reduced rates and increased workload for translators, changing working conditions and potentially affecting job satisfaction (do **Carmo2020**; **SakamotoEtAl2024**).

Ensuring the employability of graduates demands training that is aware and responsive to the deployment of AI (**EMT2022**), but education requires a critical evaluation of these tools and production models to equip students with the knowledge and skills to evaluate the impact of societal, professional and ecological sustainability (**MoorkensEtAl2024**). These concerns resonate with broader discussions about the ethical implications of AI development, including issues of bias in training data (**AkaEtAl2021**; **JohnsonEtAl2022**), the environmental impact of LLMs (**LuccioniEtAl2023**), and the risk of perpetuating cultural hegemonies through automation and standardisation (**BenderEtAl2021**).

Ensuring that students are aware of the capabilities and limitations of the systems is also an ethical task as training programmes have a responsibility to prepare students for the requirements of the market. LSCs are steadily implementing AI solutions into their workflows (**Slator2024**) and graduates need hands-on experience with these tools to successfully integrate into the profession. They need to be able to assess when the systems are suitable or not for a task. This will only be achieved through familiarisation. Further, when the systems are deemed suitable, it should also be future trained professionals who decide how to deploy them to ensure they are augmenting, not hindering, the capabilities of other professionals (O'**Brien2024**). Activities should foster critical discussions about the impact of AI on both the profession and society at large, encouraging students to consider how these tools can support rather than replace human expertise and develop the practical competencies the market demands. The goal is to prepare students to engage with AI tools ethically and strategically, understanding both their potential and limitations, while developing the agency to advocate for fair working conditions and responsible AI implementation in an increasingly automated industry.

6 Concluding remarks

Training subtitlers in the age of GenAI requires balancing subtitling-specific skills with awareness about the market and understanding media localisation practices. This type of training encompasses, firstly, ensuring graduates develop the skills they need to enter the industry demands a thorough understanding of subtitling principles, workflows and tools. Secondly, the increasing implementation of automated solutions to educate reflective professionals who can adapt to rapidly changing production settings. And lastly, training that fosters ethical awareness to support responsible engagement with technology and advocates for sustainable professional practices.

Future-proofing subtitling education requires integrating emerging technologies while maintaining core professional competencies. The problem-solving approach presented in this chapter responds to these needs by creating spaces for students to explore technological possibilities and develop foundational knowledge. Engaging with LLMs and other GenAI tools allows students to understand the potential and limitations of automation. By comparing different solutions and evaluating their suitability for specific tasks, students learn to make informed decisions about implementing automated solutions. Further, this approach encourages them to think strategically about their role in an evolving industry.

The activities discussed demonstrate how subtitling training can integrate technological developments maintaining a focus on quality and professional standards. These activities show students how to harness GenAI tools to support their work rather than allowing automation to dictate their practice. The goal is to prepare adaptable professionals who can navigate technological changes, maintain high standards and advocate for sustainable working conditions.

The continuous development of GenAI tools suggests the media localisation industry will keep evolving and graduates need to be ready to assess, critique and deploy GenAI-powered solutions. This requires careful consideration of how automation affects subtitling processes, fostering critical engagement with emerging technologies and promoting ethical awareness. Only by addressing these aspects can training programmes prepare professionals who will contribute to developing a sustainable media localisation industry.